

第二章 达朗贝尔原理 § 2.1 约束

1. 状态: $\vec{r}_i, \dot{\vec{r}}_i$

约束: 对状态之限制 $f(\vec{r}_i, \dot{\vec{r}}_i, t) = 0$

几何约束: $f(\vec{r}_i, t) = 0$

eg. 刚性杆中 $(\vec{r}_i - \vec{r}_j)^2 - r_{ij}^2 = 0$

运动约束 / 微分约束 / 速度约束

eg. 纯滚: $\dot{x} - R\dot{\theta} = 0$

转化: 对几何约束求导: $\sum_i \frac{\partial f}{\partial \vec{r}_i} \dot{\vec{r}}_i + \frac{\partial f}{\partial t} = 0$

积分 $\dot{x} - R\dot{\theta} = 0 \Rightarrow x - R\theta + C = 0$

完整 / 非完整 $\sim$ (运动约束) 可积 / 不可积

几何 / 可积运动 - 可积运动 $\sim$ (系统) 仅受完整约束 / 不仅

$\wedge \mathbb{R}$

eg. 非 $\sim$ ① $\dot{x}_0 - R\dot{\psi} = 0$

$$\Rightarrow \begin{cases} \dot{x}_0 - R\dot{\psi} \cos\varphi = 0 \\ \dot{y}_0 - R\dot{\psi} \sin\varphi = 0 \end{cases}$$

$$\begin{matrix} \downarrow & R & \downarrow \\ \varphi & & \psi \end{matrix} \quad \text{or} \quad \begin{matrix} \downarrow & & \downarrow \\ x & & y \end{matrix}$$

$x \quad y$

② $r_1 = r_2$

$$\begin{cases} (x_2 - x_1)^2 + (y_2 - y_1)^2 = l^2 \\ \frac{\dot{x}_1 + \dot{x}_2}{\dot{y}_1 + \dot{y}_2} = \frac{x_1 - x_2}{y_1 - y_2} \quad \text{质心速度沿连线} \end{cases}$$

定常 / 非定常 $\sim$ 含 t / 不含 t

单侧 / 双侧 $\sim$ 不等式 / 方程

eg. $f(\vec{r}_i, \dot{\vec{r}}_i, t) \leq 0$

2. 约束力 $\rightarrow N_i$

主动力 (其它力) $\rightarrow F_i$

3. 例: 蒸汽离心调速器.

若各杆均为轻臂, m, l, k, a, p 已知. 不转时, 弹簧原长
求 $\omega \sim \varphi$ 关系

CE, DF 张力 R_1 , DE 给球力为 R_2 , DE 给杆 R_3, R_4

杆给 AD 力 R_1, R_0

$$\text{CD: } 2R_1 \cos \varphi - p = 0 \quad (1)$$

$$\text{下杆: } R_1 = R_2 \quad (2)$$

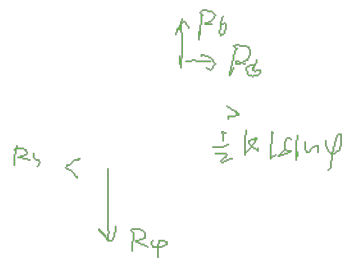
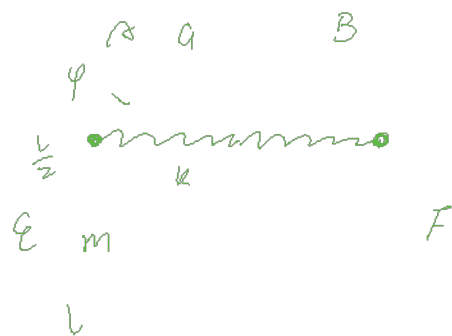
$$\text{球: } R_3 + R_2 \sin \varphi - m\omega^2(a + l \sin \varphi) = 0 \quad (3)$$

$$R_4 - R_2 \cos \varphi - mg = 0 \quad (4)$$

$$\text{上杆: } R_0 - R_4 = 0 \quad (5)$$

$$R_5 + K l \sin \varphi - R_3 = 0 \quad (6)$$

$$\text{以 A 为支点: } K l \sin \varphi \cdot \left(\frac{1}{2} l \cos \varphi\right) + R_0 l \sin \varphi - R_3 l \cos \varphi = 0 \quad (7)$$



§ 2.2 自由度和广义坐标

如果系统存在 m 个完整约束: $f_i(u_1, \dots, u_n, t) = 0$

剩下 $s = n - m$ 个独立坐标 $\rightarrow$ 有限运动中自由度 也可用参数 q 表示

即 $u_i = u_i(q_1, \dots, q_s, t)$: 坐标变换关系

独立有 $\left\{ \begin{array}{l} q_1, \dots, q_s : \text{广义坐标} \\ q_1, \dots, q_s : \text{广义速度} \end{array} \right.$, $q_s = q_s(t)$, 可以唯一地确定系统位形
不用再代入约束方程

如存在 k 个非完整约束

$$f_j(u_1, \dots, u_n, t) = 0$$

可以减少 q_s (无限小运动中的) 自由度的数量

系统的自由度 := 独立坐标无限小变化的个数

例如 直角坐标数: $3N$, 广义坐标数: $3N - m$, 自由度: $3N - m - k$

从直角坐标 $\rightarrow$ 广义坐标, 只需考虑非完整约束:

$$\text{线性的速度约束 } \sum_i A_{ji}(u_1, \dots, u_n, t) \dot{u}_i + A_j(u_1, \dots, u_n, t) = 0$$

$$\text{其中 } u_i = u_i(q_1, \dots, q_s, t) \quad \dot{u}_i = \sum_\alpha \frac{\partial u_i}{\partial q_\alpha} \dot{q}_\alpha + \frac{\partial u_i}{\partial t}$$

$$\text{代入: } \sum_i A_{ji}(u_1, \dots, u_n, t) \left(\sum_\alpha \frac{\partial u_i}{\partial q_\alpha} \dot{q}_\alpha + \frac{\partial u_i}{\partial t} \right) + A_j(u_1, \dots, u_n, t) = 0$$

$$= \sum_\alpha \left[\sum_i A_{ji}(u_1, \dots, u_n, t) \frac{\partial u_i}{\partial q_\alpha} \right] \dot{q}_\alpha + \left[\sum_i A_{ji}(u_1, \dots, u_n, t) \frac{\partial u_i}{\partial t} + A_j(u_1, \dots, u_n, t) \right]$$

$$= \sum_\alpha B_{j\alpha} \dot{q}_\alpha + B_j = 0$$

$\Rightarrow$ 在直角坐标下对速度的线性约束, 仍是广义速度的约束方程,

$$\text{一般地: } f(u_1, \dots, u_n; \dot{u}_1, \dots, \dot{u}_n; t)_{u \rightarrow q} = g(q_1, \dots, q_s; \dot{q}_1, \dots, \dot{q}_s; t)$$

$$\text{通常 } \dot{u}_i = \sum_\alpha \frac{\partial u_i}{\partial q_\alpha} \dot{q}_\alpha + \frac{\partial u_i}{\partial t} \quad \text{不会出 } \ddot{q}$$

§2.3. 虚功原理 达朗贝尔原理

1. 实位移: 满足约束和运动定律的微小位移, 实际发生, 经历时间

虚位移: 想象的, 符合约束的 δr

* 函数 $q(t)$ 微分 $dq = \dot{q} dt$

变分 $\delta q(t) = \bar{q}(t) - q(t)$

例 $q(t) = at^2$ $\bar{q}(t) = (a + \delta a)t^2$

$$dq = 2at dt \quad \delta q(t) = (\delta a)t^2 = \frac{\partial q}{\partial a} \delta a$$

$$q(t) = at^\alpha \quad (a \text{ 为参}) \quad dq = da t^{\alpha-1}$$

$$\delta q(t) = a(t^{\alpha+\delta\alpha} - t^\alpha) = at^\alpha (e^{\delta\alpha} - 1) = \frac{\partial q}{\partial a} \delta a$$

$$= at^\alpha (\delta\alpha) \ln t$$

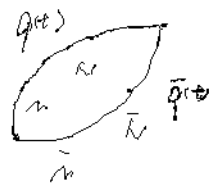
*

$$1. M \rightarrow N \rightarrow \bar{N}$$

$$q(t) \rightarrow q(t+dt) = q(t) + dq(t)$$

$$\Rightarrow \bar{q}(t+dt) = q(t+dt) + \delta q(t+dt)$$

$$= q(t) + dq(t) + \delta q(t) + \delta dq(t)$$



$$2. M \rightarrow \bar{M} \rightarrow \bar{N}$$

$$q(t) \rightarrow \bar{q}(t) = q(t) + \delta q(t)$$

$$\Rightarrow \bar{q}(t+dt) = \bar{q}(t) + d\bar{q}(t) = q(t) + \delta q(t) + dq(t) + d\delta q(t)$$

$$\Rightarrow \text{在等时条件下} \quad \delta q(t) = d\delta q(t)$$

$$\delta q = 0$$

实位移: $\vec{r} \rightarrow \vec{r} + d\vec{r}$
 $t \rightarrow t + dt$

$$\begin{cases} f(\vec{r}, t) = 0 \\ f(\vec{r} + d\vec{r}, t + dt) = 0 \end{cases} \Rightarrow df(\vec{r}, t) = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz + \frac{\partial f}{\partial t} dt = 0$$

$$= \nabla f \cdot \underbrace{d\vec{r}}_{\text{实位移}} + \frac{\partial f}{\partial t} dt = 0$$

+ 动力学方程

同一轨道上不同时间

满足动力学方程且唯一

虚位移: $\vec{r} \rightarrow \vec{r} + \delta\vec{r}$
 t

$$\begin{cases} f(\vec{r}, t) = 0 \\ f(\vec{r} + \delta\vec{r}, t) = 0 \end{cases} \quad \delta f = \frac{\partial f}{\partial x} \delta x + \frac{\partial f}{\partial y} \delta y + \frac{\partial f}{\partial z} \delta z = \nabla f \cdot \delta\vec{r} = 0$$

(等时变化)

连接不同轨道的同一时刻

不满足动力学方程且无限多

几何中: 张成约束曲面的切平面

2. 虚功: $\delta W = \vec{F} \cdot \delta\vec{r}$

约束: $f(\vec{r}, t) = 0$

约束力的虚功

约束力: $\vec{N} = \lambda \nabla f(\vec{r}, t) \Rightarrow \delta W = \sum_i \vec{N}_i \cdot \delta\vec{r}_i = 0 \rightarrow$ 理想约束

* 实功: $dW = \vec{N} \cdot d\vec{r} = \lambda \nabla f(\vec{r}, t) \cdot d\vec{r} = \lambda df = \lambda \frac{\partial f}{\partial t} dt = -\lambda \frac{\partial f}{\partial t} dt$

若约束 f 是稳定的, 则 $\frac{\partial f}{\partial t} = 0 \Rightarrow$ 实功 $dW = 0$

例: 静止光滑曲面 $f(x, y, z) = 0$

例: 刚体在曲面上作无滑滚动

$\vec{N} = \lambda \nabla f = \lambda \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$

接触点 P : $\vec{v}_P = 0 \Rightarrow \delta\vec{r}_P = 0$

$(\vec{N} + \vec{F}) \cdot \delta\vec{r}_P = 0$

$$\delta W = N \cdot \delta r = \lambda \left(\frac{\partial f}{\partial x} \delta x + \frac{\partial f}{\partial y} \delta y + \frac{\partial f}{\partial z} \delta z \right) = 0$$

$$f(x, y, z, t) = 0$$

$$d\vec{r} = (dx, dy, dz)$$

$$\delta W = N \cdot \delta \vec{r} = 0, \quad dW = N \cdot d\vec{r} = \lambda \nabla f \cdot d\vec{r} \neq 0$$

虚功原理：一个只有理想约束的力学系统处于平衡状态，则系统所有主动力

虚功之和为0

证：平衡状态： $\vec{F}_i + \vec{N}_i = 0$

• $\delta W_i = \vec{F}_i \cdot \delta \vec{r}_i + \vec{N}_i \cdot \delta \vec{r}_i = 0$ 又，理想： $\vec{N}_i \cdot \delta \vec{r}_i = 0$

• $\delta W = \sum_i \vec{F}_i \cdot \delta \vec{r}_i = 0$ 若有非理想约束，将非理想约束反力包括进主动力即可

3. 广义坐标下： $r_i = r_i(q, \dots, t)$

$$\delta r_i = \sum_j \frac{\partial r_i}{\partial q_j} \delta q_j$$

$$\delta W = \sum_i \vec{F}_i \cdot \left(\sum_j \frac{\partial r_i}{\partial q_j} \delta q_j \right)$$

$$= \sum_j \left(\sum_i \vec{F}_i \cdot \frac{\partial r_i}{\partial q_j} \right) \delta q_j = 0$$

令 $Q_j = \sum_i \vec{F}_i \cdot \frac{\partial r_i}{\partial q_j}$ 称为广义力

• $\delta W = \sum_j Q_j \delta q_j = 0$

(q_j 相互独立)

对完整约束： $\Rightarrow Q_j = 0$

例：已知 m_1, m_2, l_1, l_2, F, g 求 θ_1, θ_2

主动力： $F, m_1 g, m_2 g \Rightarrow m_1 g \delta x_1 + m_2 g \delta x_2 + F \delta y_3 = 0$

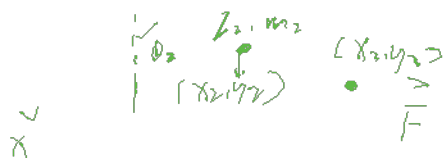
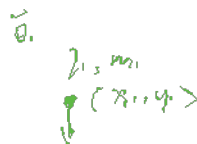
利用广义坐标： θ_1, θ_2

$$x_1 = \frac{1}{2} l_1 \cos \theta_1, \quad x_2 = l_1 \cos \theta_1 + \frac{1}{2} l_2 \cos \theta_2$$

$$y_3 = l_1 \sin \theta_1 + l_2 \sin \theta_2 = 0$$

$\Rightarrow m_1 g \left(-\frac{1}{2} l_1 \sin \theta_1 \delta \theta_1 \right) + m_2 g \left(-l_1 \sin \theta_1 \delta \theta_1 - \frac{1}{2} l_2 \sin \theta_2 \delta \theta_2 \right)$

$+ F \left(l_1 \cos \theta_1 \delta \theta_1 + l_2 \cos \theta_2 \delta \theta_2 \right) = 0$



$$\Rightarrow \begin{cases} -\frac{1}{2}m_1 g l_1 \sin \theta_1 - m_2 g l_1 \sin \theta_1 + F l_1 \cos \theta_1 = 0 \\ -\frac{1}{2}m_2 g l_2 \sin \theta_2 + F l_2 \cos \theta_2 = 0 \end{cases} \Rightarrow \begin{cases} \tan \theta_1 = \frac{l_2}{l_1} \\ \tan \theta_2 = \frac{2F l_2}{m_2 g l_2} \end{cases}$$

例: 已知 1. α , 求平衡时 θ (A 处不光滑)

$$A(-a, -a \cot \theta)$$

设 A 处约束反力为 N_A

$$C(\frac{1}{2} \sin \theta - a, \frac{1}{2} \cos \theta - a \cot \theta)$$



$$\delta W = N_A \delta y_A + mg \cdot \delta y_C = mg \cdot \delta y_C$$

$$= mg \left(-\frac{1}{2} l \sin \theta + a \frac{\cos^2 \theta}{\tan^2 \theta} \right) \delta \theta = 0$$

$$\Rightarrow \sin \theta = \left(\frac{2a}{l} \right)^{\frac{1}{2}}$$

4. 主动力为保守力时.

$$\delta W = -\delta V = \sum_j Q_j \delta q_j$$

$$\delta W = \sum_i F_i \delta x_i = \sum_j F_i \left(\sum_j \frac{\partial x_i}{\partial q_j} \right) \delta q_j$$

$$= \sum_j \left(\sum_i F_i \frac{\partial x_i}{\partial q_j} \right) \delta q_j \quad \text{代入 } F_i = -\frac{\partial V}{\partial x_i}$$

$$= \sum_j -\frac{\partial V}{\partial q_j} \delta q_j \rightarrow Q_j = -\frac{\partial V}{\partial q_j} \quad \text{理想约束, 平衡时} \rightarrow Q_j = 0$$

例: 已知 m_1, m_2, l_1, l_2, F, g , 求 θ_1, θ_2

* 使用广义力

$$r_1 = \frac{l_1}{2} (\cos \theta_1 \hat{x} + \sin \theta_1 \hat{y})$$

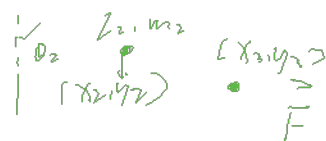
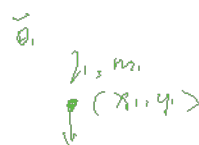
$$r_2 = (l_1 \cos \theta_1 + \frac{1}{2} l_2 \cos \theta_2) \hat{x} + (l_1 \sin \theta_1 + \frac{1}{2} l_2 \sin \theta_2) \hat{y}$$

$$r_3 = (l_1 \cos \theta_1 + l_2 \cos \theta_2) \hat{x} + (l_1 \sin \theta_1 + l_2 \sin \theta_2) \hat{y}$$

$$Q_1 = m_1 g \cdot \frac{\partial r_1}{\partial \theta_1} + m_2 g \cdot \frac{\partial r_2}{\partial \theta_1} + F \cdot \frac{\partial r_3}{\partial \theta_1}$$

$$= -\frac{1}{2} l_1 (\sin \theta_1 \hat{x} - \cos \theta_1 \hat{y}) m_1 g + [-l_1 \sin \theta_1 \hat{x} + l_1 \cos \theta_1 \hat{y}] m_2 g + [-l_1 \sin \theta_1 \hat{x} + l_1 \cos \theta_1 \hat{y}] F$$

$$Q_2 = m_1 g \cdot \frac{\partial r_1}{\partial \theta_2} + m_2 g \cdot \frac{\partial r_2}{\partial \theta_2} + F \cdot \frac{\partial r_3}{\partial \theta_2}$$



$$= \frac{1}{2} l_2 (-\sin \theta_2 \hat{x} + \cos \theta_2 \hat{y}) m_2 g + l_1 (-\sin \theta_2 \hat{x} + \cos \theta_2 \hat{y}) F$$

$$\delta W = Q_1 \delta \theta_1 + Q_2 \delta \theta_2 = 0 \quad \because \delta \theta_1, \delta \theta_2 \text{ 相互独立} \therefore Q_1 = 0, Q_2 = 0$$

$$\Rightarrow \tan \theta_1 = \underline{\hspace{2cm}} \quad \tan \theta_2 = \underline{\hspace{2cm}}$$

$$\begin{aligned} \text{* 使用势能: } V &= -m_1 g \cdot \frac{1}{2} l_1 \cos \theta_1 + (-m_2 g) (l_1 \cos \theta_1 + \frac{1}{2} l_2 \cos \theta_2) \\ &\quad + F \cdot (l_1 \sin \theta_1 + l_2 \sin \theta_2) \end{aligned}$$

$$\begin{cases} \frac{\partial V}{\partial \theta_1} = 0 \\ \frac{\partial V}{\partial \theta_2} = 0 \end{cases} \Rightarrow \begin{cases} \theta_1 = 0 \\ \theta_2 = 0 \end{cases} \rightarrow \begin{aligned} \tan \theta_1 &= \underline{\hspace{2cm}} \\ \tan \theta_2 &= \underline{\hspace{2cm}} \end{aligned}$$

例: 已知 L, R, m_1, m_2 , 求 θ

设 $\beta, (x_1, y_1), (x_2, y_2)$, 右绳长 L

坐标变换关系: (只与主动力方向平行或垂直)

$$\begin{cases} y_1 = L \cos \theta \\ y_2 = L \cos \theta + (L - R\beta - \sqrt{L^2 - R^2}) \end{cases}$$

$$\begin{aligned} \hookrightarrow \begin{cases} \delta y_1 = -L \sin \theta \delta \theta \\ \delta y_2 = -L \sin \theta \delta \theta - R \delta \beta \end{cases} \end{aligned}$$

$$\text{约束: } L \sin(\theta + \beta) = R \rightarrow (\cos \theta + \beta) (\delta \theta + \delta \beta) = 0$$

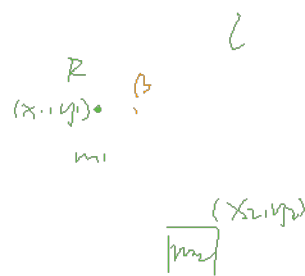
$$\hookrightarrow \begin{cases} \delta y_1 = -L \sin \theta \delta \theta \\ \delta y_2 = (-L \sin \theta + R) \delta \theta \end{cases}$$

$$\text{虚功原理: } \left[m_1 g L \sin \theta + m_2 g (R - L \sin \theta) \right] \delta \theta = 0$$

$$\text{解得 } \theta = \arcsin \left(\frac{m_2 R}{(m_1 + m_2) L} \right)$$

$$\text{或 } V(y_1, y_2) = V'(\theta)$$

$$\begin{aligned} &\theta \\ &L - R - \beta \end{aligned}$$



$$\delta \vec{r} = 0 \rightarrow \delta = \dots$$

5. 拉格朗日未定乘数 (求出约束力)

n 个质点, k 个约束: $f_\beta(x, y, z) = 0, \beta = 1, \dots, k$. 注: 此时 q_α (δq_α) 不独立
故取 $Q_\alpha = 0$

① $3n$ 个坐标求解 ($3n-k$ 个坐标独立) ② 用广义坐标 $q_\alpha, \alpha = 1, \dots, s$

虚功原理: $\sum_i \vec{F}_i \cdot \delta \vec{r}_i = 0$ 虚功原理: $\delta W = \sum_\alpha Q_\alpha \delta q_\alpha = 0$

虚位移: $\sum_i \delta f_{\beta,i} = \sum_i \nabla f_{\beta,i} \cdot \delta \vec{r}_i = 0 \Leftrightarrow$ 虚位移: 约束 $\varphi_\beta(q_1, \dots, q_s) = 0, \beta = 1, \dots, k$

$$\Rightarrow \sum_i (\vec{F}_i + \sum_\beta \lambda_\beta \nabla f_{\beta,i}) \cdot \delta \vec{r}_i = 0$$

$$\text{分量: } x: \sum_i (F_{ix} + \sum_\beta \lambda_\beta \frac{\partial f_{\beta,i}}{\partial x_i}) \delta x_i = 0$$

$$y: \dots$$

$$z: \dots$$

} 3n 个

选取 k 个 $Q_\beta \rightarrow F_{ix} + \sum_\beta \lambda_\beta \frac{\partial f_{\beta,i}}{\partial x_i} = 0$

即 $\sum_{k+1} (F_{ix} + \sum_\beta \lambda_\beta \frac{\partial f_{\beta,i}}{\partial x_i}) \delta x_i = 0$

剩下 $3n-k$ 个 $F_{ix} + \sum_\beta \lambda_\beta \frac{\partial f_{\beta,i}}{\partial x_i}$ 也必为零

(因为一维独立)

$\Rightarrow$ 共有 $3n+k$ 个方程确定 $3n$ 个 x_i, y_i, z_i

和 k 个 λ_β

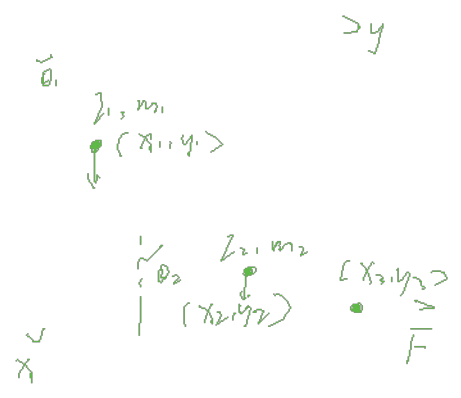
例: 求两杆之间的相互作用力

选取 Q_1, Q_2, x_2, y_2 要求解约束力, 就不能只选取自由度数度的广义坐标

主动力的虚功原理:

$$m_1 g \delta x_1 + m_2 g \delta x_2 + F \delta y_2$$

$$= -\frac{1}{2} m_1 g l_1 \sin \theta_1 \delta \theta_1 + m_2 g \delta x_2 + \frac{1}{2} m_2 g l_2 \sin \theta_2 \delta \theta_2 + F \delta y_2$$



$$= 0$$

约束方程:

$$\begin{cases} x_2 - l_1 \sin \theta_1 - l_2 \sin \theta_2 = 0 \\ y_2 - l_1 \cos \theta_1 - l_2 \cos \theta_2 = 0 \end{cases}$$

虚功原理: $-\frac{1}{2} m_1 g l_1 \sin \theta_1 \delta \theta_1 + m_2 g \delta x_2 + \frac{1}{2} F l_2 \cos \theta_2 \delta \theta_2 + F \delta y_2$

$$+ \lambda_1 \delta (x_2 - l_1 \sin \theta_1 - l_2 \sin \theta_2) + \lambda_2 \delta (y_2 - l_1 \cos \theta_1 - l_2 \cos \theta_2)$$

$$= -\frac{1}{2} m_1 g l_1 \sin \theta_1 \delta \theta_1 + m_2 g \delta x_2 + \frac{1}{2} F l_2 \cos \theta_2 \delta \theta_2 + F \delta y_2$$

$$+ \lambda_1 \delta x_2 - \lambda_1 l_1 \cos \theta_1 \delta \theta_1 - \lambda_1 l_2 \cos \theta_2 \delta \theta_2 + \lambda_2 \delta y_2 + \lambda_2 l_1 \sin \theta_1 \delta \theta_1 + \lambda_2 l_2 \sin \theta_2 \delta \theta_2$$

$$= (m_2 g + \lambda_1) \delta x_2 + (F + \lambda_2) \delta y_2 - l_1 \left(\frac{1}{2} m_1 g \sin \theta_1 + \lambda_1 \cos \theta_1 - \lambda_2 \sin \theta_1 \right) \delta \theta_1$$

$$+ l_2 \left(\frac{1}{2} F \cos \theta_2 - \lambda_1 \cos \theta_2 + \lambda_2 \sin \theta_2 \right) \delta \theta_2 = 0$$

求解约束: $\lambda_1 = -m_2 g, \lambda_2 = -F$

求解角度: $\tan \theta_1 = \frac{m_2 g}{\frac{1}{2} m_1 g + F}, \tan \theta_2 = \frac{\frac{1}{2} F + m_2 g}{F}$

b. 达朗贝尔原理.

$$\sum_i (\vec{F}_i + \vec{N}_i - m_i \vec{\ddot{r}}_i) = 0$$

在理想约束下: 虚功原理 $\sum_i (\vec{F}_i - m_i \vec{\ddot{r}}_i) \cdot \delta \vec{r}_i = 0$

例: $x_c = a + l \sin \varphi$, $\delta x_c = l \cos \varphi \delta \varphi$

$$y_c = 2l \cos \varphi \quad \delta y_c = -2l \sin \varphi \delta \varphi$$

$$x_a = a + \frac{1}{2} l \sin \varphi \quad \delta x_a = \frac{1}{2} l \cos \varphi \delta \varphi$$

$$y_a = l \cos \varphi, \quad \delta y_a = -l \sin \varphi \delta \varphi$$

达朗贝尔原理:

$$2m\omega^2 x_c \delta x_c + P \delta y_c + [-2kx_a - 0] 2\delta x_a + 2mg \delta y_a = 0$$

$$2m\omega^2 (a + l \sin \varphi) l \cos \varphi \delta \varphi - 2Pl \sin \varphi \delta \varphi - kl \sin \varphi (l \cos \varphi \delta \varphi) - 2mg l \sin \varphi \delta \varphi = 0$$

$$2m\omega^2 l \cos \varphi (a + l \sin \varphi) - 2Pl \sin \varphi - kl^2 \sin \varphi \cos \varphi - 2mg l \sin \varphi = 0$$

$$\omega^2 = \frac{2P + 2mg + kl \cos \varphi}{2m\omega^2 (a + l \sin \varphi)} \tan \varphi$$

